WORKSHEET

Chapter 10: Margins of error for standard normal variables

1 CAS Find the confidence levels corresponding to the following margins of error in standard normal variables.

a $E = 2$

b $E = 1.8$

c $E = 0.4$

d $E = 0.25$

2 CAS Find the margins of error needed in standard normal variables to give the following confidence levels.

a $a = 0.96$

b $a = 0.65$

c 90% confidence

d 75% confidence

3 CAS Find the margins of error needed in standard normal variables to give the following confidence levels.

a $a = 0.85$

b $a = 0.996$

c 77% confidence

d 99.9% confidence

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4 CAS Find the confidence intervals in standard normal variables needed to give the following confidence levels.

a $a = 0.92$

b $a = 0.89$

c 74% confidence

d 97% confidence

5 CAS Find the confidence levels corresponding to the following confidence intervals in standard normal variables.

a $-2 \leq z \leq 2$

b $-1.8 \leq z \leq 1.8$

c $-2.4 \leq z \leq 2.4$

d $-2.8 \leq z \leq 2.8$

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Answers

1 **a** 95%

b 93%

c 31%

d 20%

2 **a** 2.05

b 0.934

c 1.64

d 1.15

3 **a** 1.44

b 2.88

c 1.20

d 3.29

4 **a** $-1.75 < z < 1.75$

b $-1.60 < z < 1.60$

c $-1.13 < z < 1.13$

d $-2.17 < z < 2.17$

5 **a** 95%

b 93%

c 98%

d 99%

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